

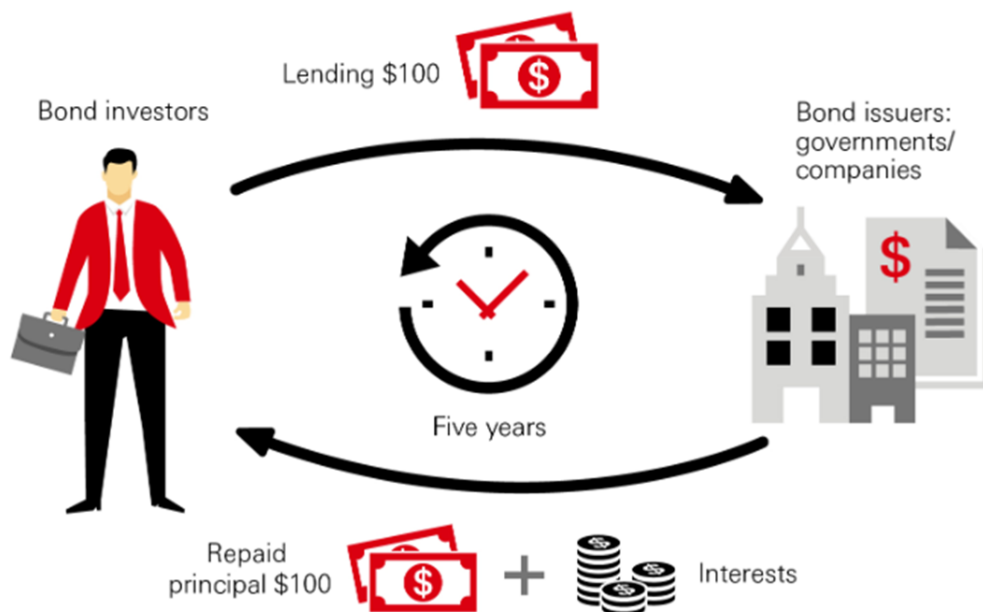
Introduction to Fixed Income Investment



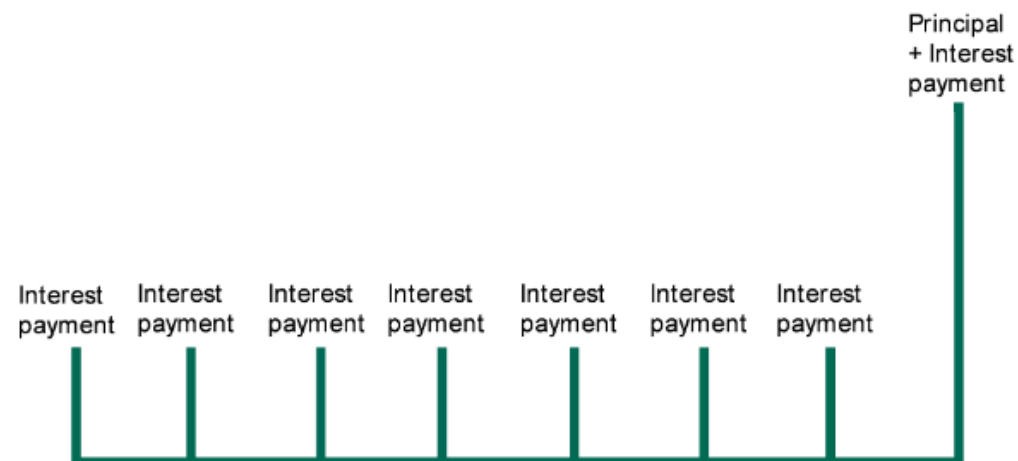
ตราสารหนี้ คือ ตราสารทางการเงินที่ผู้ถือ (นักลงทุน) มีสถานะเป็นเจ้าหนี้ และผู้ออกมีสถานะเป็นลูกหนี้ โดยเจ้าหนี้จะได้รับผลตอบแทนในรูปของ “ดอกเบี้ย” อย่างสม่ำเสมอตามระยะเวลาที่กำหนดไว้ และจะได้รับ “เงินต้น” คืนเมื่อครบกำหนดอายุ ตัวอย่างตราสารหนี้ที่พบเห็นทั่วไป เช่น ตั๋วเงินคลัง พันธบัตรรัฐบาล และหุ้นกู้เอกชน

Why Investing in Fixed Income?

Investing in bonds for income potential



Provide a periodic stream of cash flows



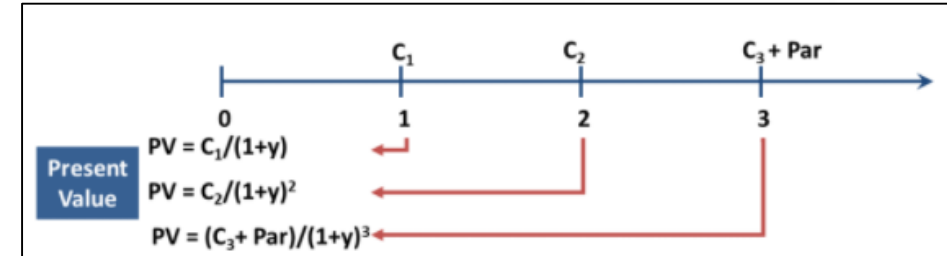
- Principal protection
- Hedge against inflation
- Diversification

- **Bond maturity:** the maturity date of a bond is the date which the principal is to be repaid.
 - **Term to maturity** (or bond tenor): the time remaining until maturity.
 - **Money market securities:** bonds with original maturities of one year or less.
- **Par value:** the principal amount of a bond that will be repaid at maturity. Also referred as face value or principal value of a bond
- **Coupon rate:** the annual percentage of its par value that will be paid to bondholders. Coupon payment of a bond can be annually, semi-annually or quarterly.
- **Credit ratings:** designations used by ratings services to give relative indications of credit quality
- **Credit spread:** a yield difference, typically in relation to a comparable U.S. Treasury security, that reflects the issuer's credit quality.
- **Default:** failure to pay principal or interest when due.
- **Duration:** the weighted maturity of a fixed-income investment's cash flows, used in the estimation of the price sensitivity of fixed-income securities for a given change in interest rates.
- **Over-the-counter market (OTC):** a securities market that is conducted by dealers throughout the country through negotiation of price rather than through the use of an auction system as represented by a stock exchange.
- **Yield to maturity:** a yield on a security calculated by assuming that interest payments will be made until the final maturity date. Yield to maturity is essentially the discount rate at which the present value of future payments (investment income and return of principal) equals the price of the security.



Coupon Bond Pricing Formula

$$\text{Bond Price} = \frac{\text{Coupon 1}}{(1 + YTM)^1} + \frac{\text{Coupon 2}}{(1 + YTM)^2} + \dots + \frac{\text{Coupon } n}{(1 + YTM)^n} + \frac{\text{Face Value}}{(1 + YTM)^n}$$



Where,

- Bond price = the current price of the bond.
- Coupon = Multiple interests received during the investment horizon. These are reinvested back at a constant rate.
- Face value = The price of the bond set by the issuer.
- YTM = the discount rate at which all the present value of bond future cash flows equals its current price.

Coupon Bond Pricing Formula

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Yield to Maturity Formula

$$\text{Yield to Maturity (YTM)} = \frac{C + \frac{FV - PV}{n}}{\frac{FV + PV}{2}}$$

Key Concept

- Yield to maturity is the **discount rate** at which the present value of future payments (investment income and return of principal) equals the price of the security. **YTM is essentially a bond's internal rate of return if held to maturity.**
- Calculating the yield to maturity assumes all coupon or interest payments can be reinvested at the same rate of return as the bond.
- An important distinction between a bond's YTM and its coupon rate is the YTM fluctuates over time based on the prevailing interest rate environment, whereas the coupon rate is fixed.
- The relationship between the yield to maturity and coupon rate are as follows;
 - Yield to Maturity (YTM) < Coupon Rate → The bond is being sold at a “premium” to its par value.
 - Yield to Maturity (YTM) > Coupon Rate → The bond is being sold at a “discount” to its par value.
 - Yield to Maturity (YTM) = Coupon Rate → The bond is said to be “trading at par”.

Bond Yield and Price Relationship



- If the bond yield rises, the bond price falls... and vice versa

If a bond is held to maturity, any price gains over the life of the bond are not realized; instead, the bond's price typically reverts to par (100) as it nears maturity and repayment of the principal.

Duration Formula

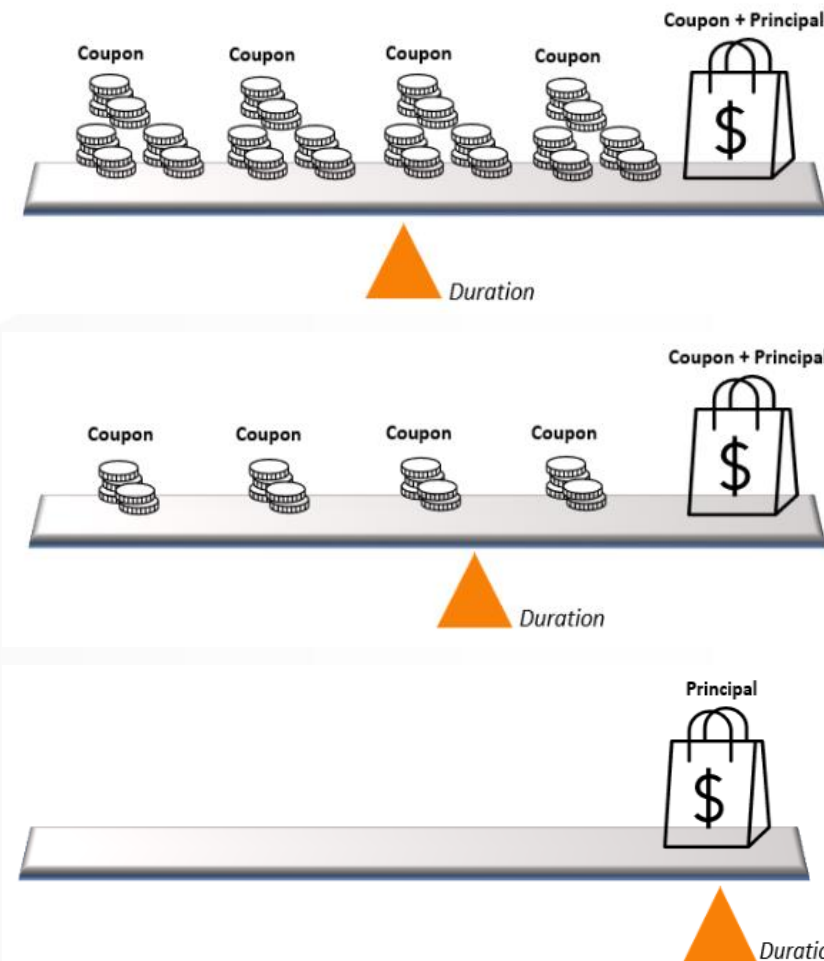
$$\text{Macaulay duration} = \sum_{i=1}^n \left(\frac{t_i \cdot C_i}{(1+y)^i} \right) + \frac{t_n \cdot (C_n + M)}{(1+y)^n}$$

Where:

- n = number of periods until each cash flow is received, usually in years
- i = number of cash flows
- t_i = time until the i th cash flow is received
- C_i = cash flow at time i
- y = yield to maturity
- M = amount of the final principal payment

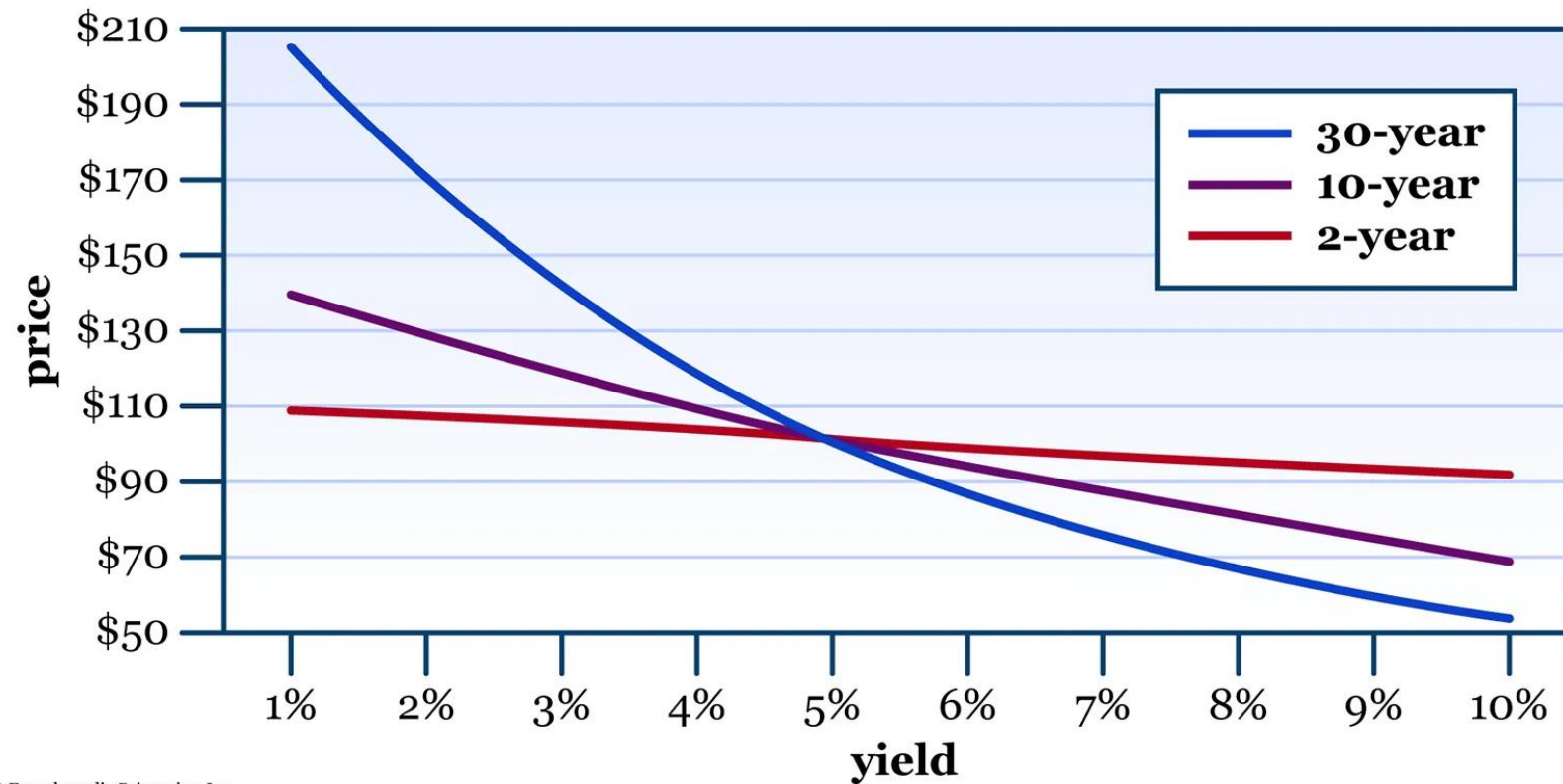
$$\text{Modified duration} = \frac{\text{Macaulay duration}}{1 + \left(\frac{\text{yield to maturity}}{\text{number of coupon payments per year}} \right)}$$

- Modified Duration tells us how much a bond's price will change with respect to a change in yield.



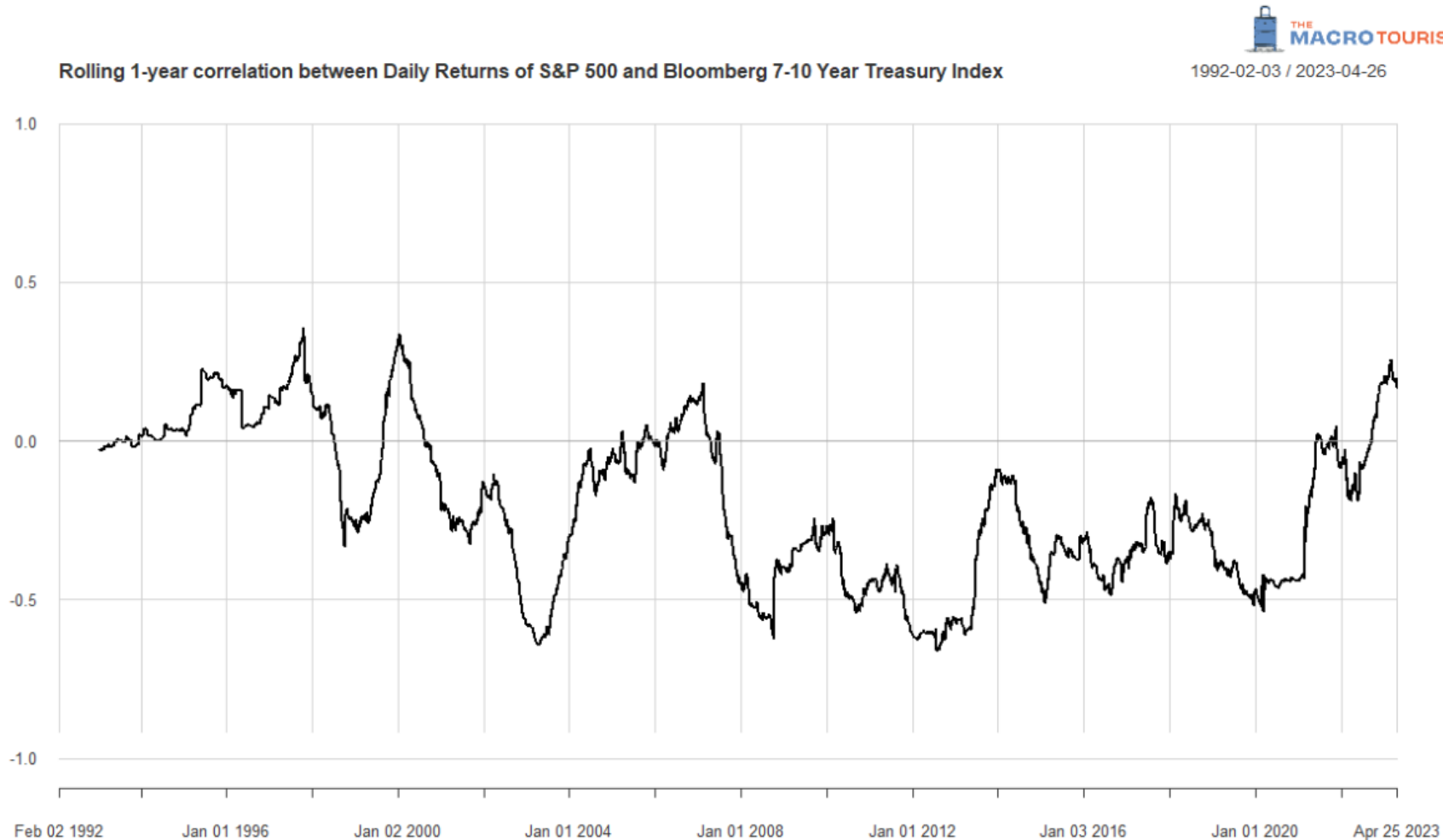
- For zero-coupon bonds, duration = time to maturity

Duration: The sensitivity of a bond's price to changes in yield



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- The higher the duration, the more sensitive of a bond price to a changing yield



- One of the main reasons for including fixed income in a multi-asset portfolio is to diversify risk.
- When equities and fixed income have a negative correlation, it means that they tend to move in opposite directions. This negative correlation can help reduce the overall volatility of the portfolio

Different type of risks

Interest rate/
duration risk



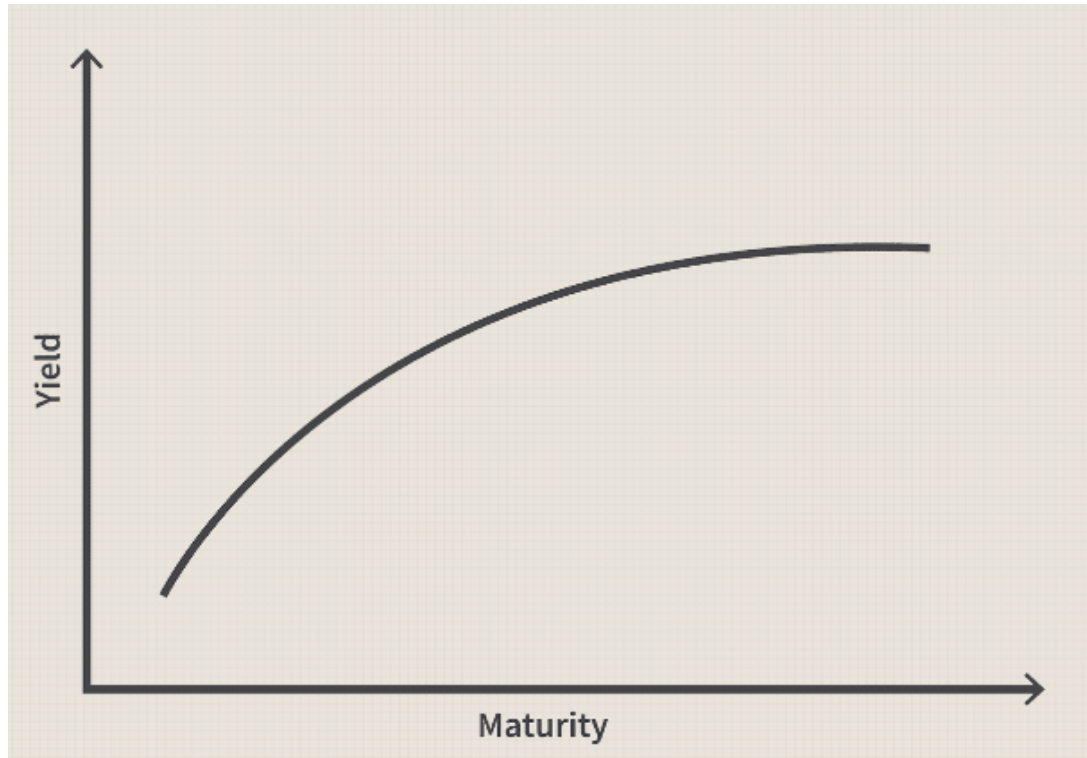
Credit/default risks



Liquidity risk



Yield Curve



What is a Yield Curve

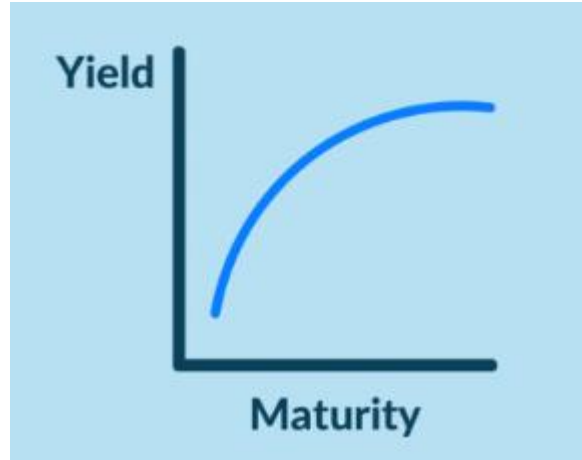
- A yield curve is a graphical line representing the interest rates on bonds on various maturities.
- The curve is plotted on a graph where a bond's yield is shown on the vertical axis while the time to maturity is shown on the horizontal axis. The curve is drawn by plotting the yields of the Bonds according to various maturity dates.
- Usually, the longer the maturities, the higher the interest rate they offer, which means an upward-sloping curve.

What Are the Factors Influencing the Yield Curve?

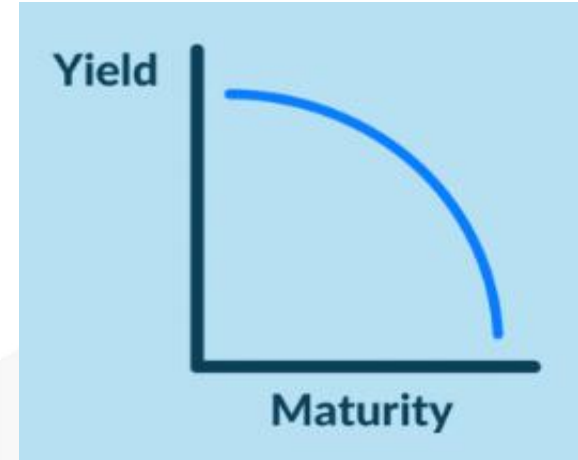
- Uncertainty risk with a longer time frame in the future
- Monetary Policy (interest rate, QE, QT)
- Inflation expectation
- Economic growth outlook

Yield Curve Shapes

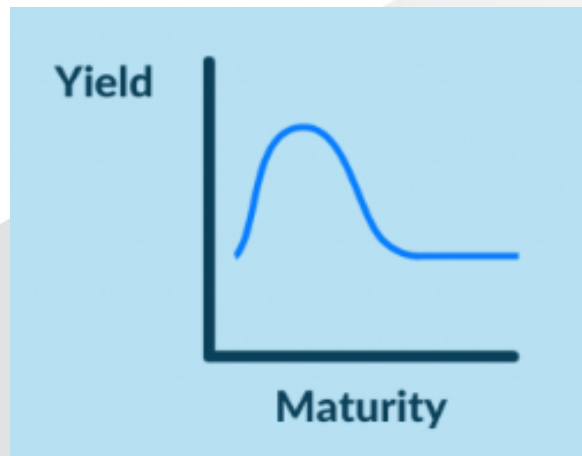
Upward Sloping or Normal Yield Curve



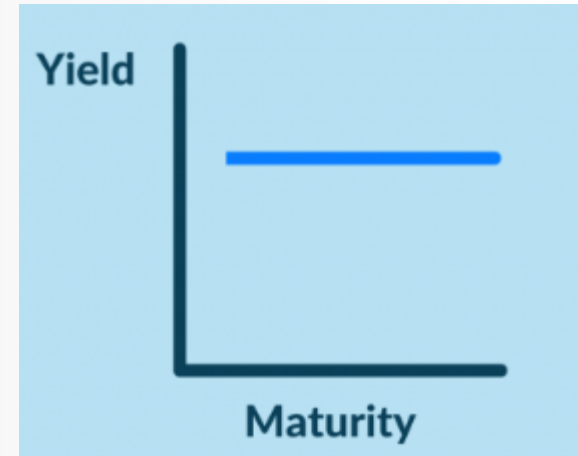
Inverted Yield Curve



Humped Yield Curve

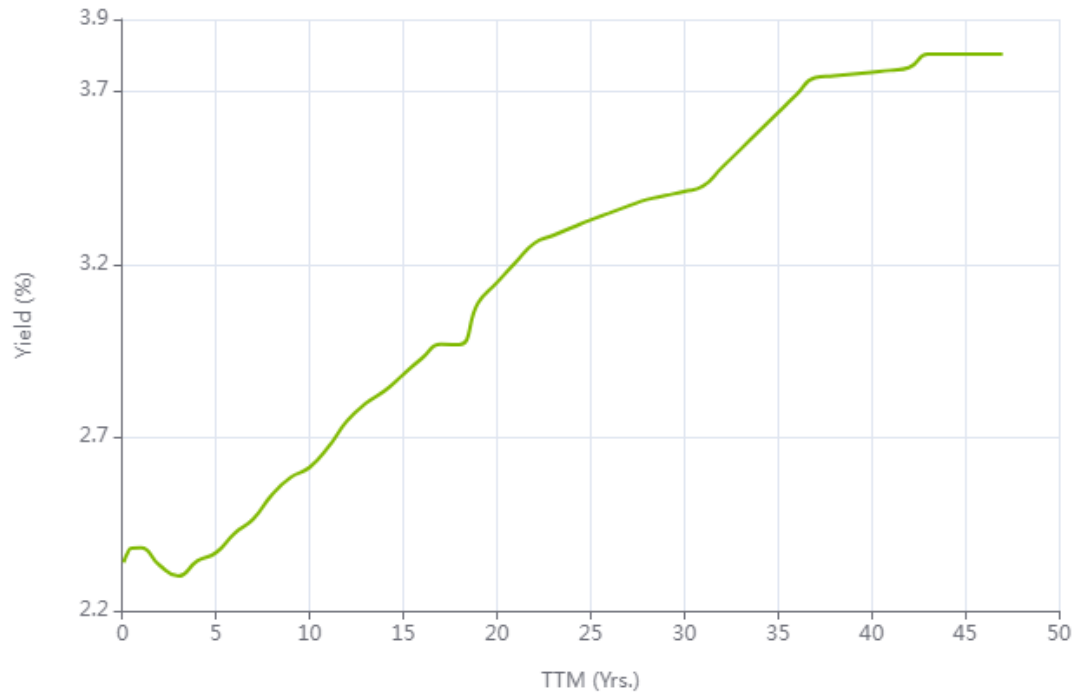


Flat Yield Curve



Thai Government Bond Yield Curve

ThaiBMA Government Bond Yield Curve
as of 26 July 2024



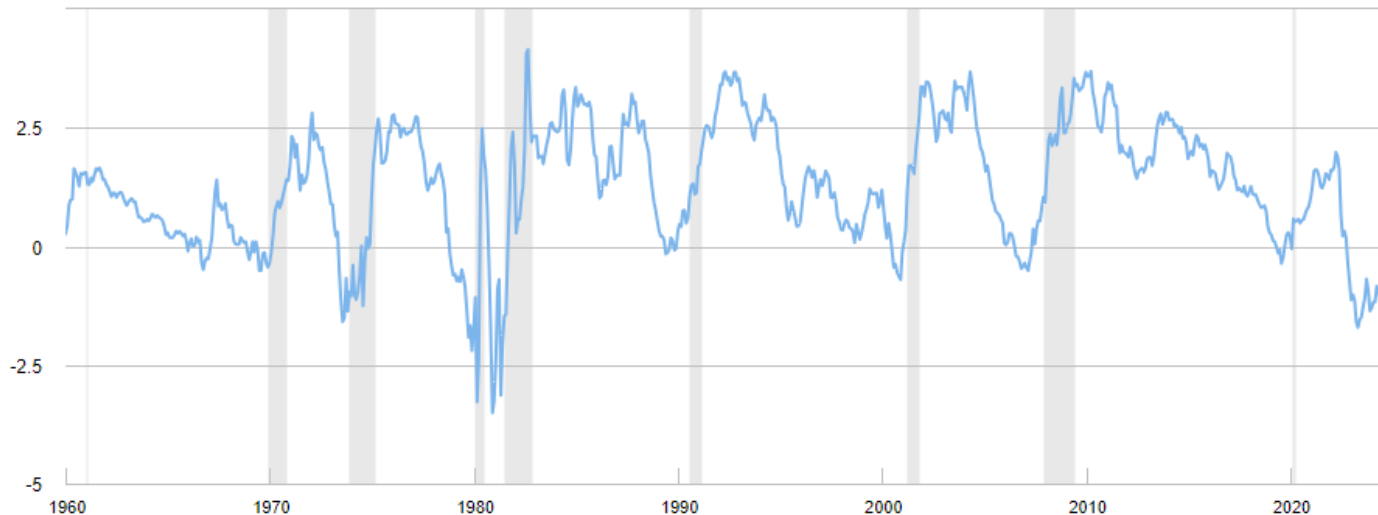
US Treasury Yield Curve



Yield Curve as a Risk Indicator

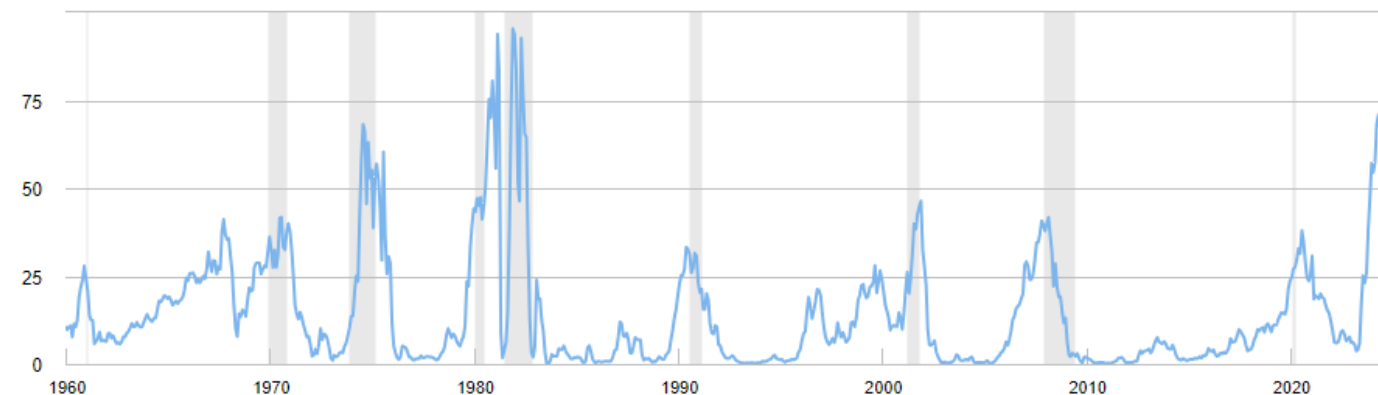
Treasury Term Spread: 10 Year Bond Rate - 3 Month Bill Rate

Percentage points (monthly average)



Probability of U.S. Recession, Twelve Months Ahead of Term Spread Readings

Percent (monthly average)



The Yield Curve as a Leading Indicator

- The bond yield curve is one of the indicators used to measure the overall risk perception of investors in the economy.
- It is commonly used as a leading economic indicator, especially when it takes on an inverted shape which is regarded as a sign of an economic downturn.

International Rating Scale

Credit risk	Bond rating	
	Moody's	Standard & Poor's
Investment grade (High grade)		
Highest quality	Aaa	AAA
High quality (very strong)	Aa	AA
Upper medium grade (strong)	A	A
Medium grade	Baa	BBB
Non-investment grade (High yield)		
Lower medium grade (somewhat speculative)	Ba	BB
Lower grade (speculative)	B	B
Poor quality (may default)	Caa	CCC
Most speculative	Ca	CC
No interest being paid or bankruptcy petition filed	C	C
In default	D	D

Domestic Rating Scale

ระดับเครดิต	TRIS & FITCH
สูงสุด	AAA
สูง	AA+
	AA
	AA-
ปานกลาง-สูง	A+
	A
	A-
ต่ำ-ปานกลาง	BBB+
	BBB
	BBB-
เก็งกำไร	BB+
	BB
	BB-
เก็งกำไรอย่างสูง	B
เสี่ยงสูงมาก	CCC+
	CCC
	CCC-
เก็งกำไรชัดเจน	C
ไม่สามารถชำระหนี้คืนได้	D

Example of Global Top Credit Rating Countries

Global Credit Ratings

	Fitch	Standard & Poors	Moody's
Germany	AAA	AAA	Aaa
Denmark	AAA	AAA	Aaa
Netherlands	AAA	AAA	Aaa
Sweden	AAA	AAA	Aaa
Norway	AAA	AAA	Aaa
Switzerland	AAA	AAA	Aaa
Luxembourg	AAA	AAA	Aaa
Singapore	AAA	AAA	Aaa
Australia	AAA	AAA	Aaa
US	AA+	AA+	Aaa

Source: Bloomberg

Note: Local currency long-term rating

Bloomberg

Thailand		Brow
91) Financial Snapshot	92) Economics	93) Risk
Agency	Rating	
Standard & Poor's	BBB+	
Moody's	Baa1	
Fitch	BBB+	

Cumulative Historical Default Rate for Global Financial Sector

Global financial services average cumulative default rates (1981- 2022) (%)

--Time horizon (years)--

Rating	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
AAA	0.00	0.04	0.21	0.38	0.55	0.72	0.80	0.89	0.93	0.97	1.02	1.06	1.10	1.15	1.19
AA	0.03	0.09	0.17	0.30	0.44	0.57	0.67	0.78	0.88	0.97	1.04	1.10	1.15	1.21	1.26
A	0.07	0.18	0.31	0.44	0.57	0.71	0.86	0.97	1.06	1.17	1.26	1.33	1.40	1.47	1.56
BBB	0.21	0.55	0.92	1.29	1.60	1.88	2.12	2.35	2.61	2.82	3.07	3.33	3.60	3.90	4.19
BB	0.59	1.45	2.35	3.17	3.93	4.68	5.24	5.61	5.93	6.20	6.54	6.83	6.92	6.96	7.02
B	2.37	4.84	6.77	8.14	9.27	10.29	10.98	11.71	12.46	13.19	13.93	14.44	14.86	15.32	15.66
CCC/C	16.89	21.85	24.58	26.68	27.97	28.63	29.56	30.77	31.29	31.87	31.87	32.20	32.89	33.66	34.08
Investment grade	0.10	0.25	0.44	0.64	0.82	1.00	1.15	1.29	1.42	1.55	1.67	1.77	1.89	2.00	2.12
Speculative grade	2.17	3.90	5.31	6.43	7.37	8.21	8.85	9.41	9.91	10.37	10.84	11.21	11.46	11.71	11.89
All rated	0.48	0.92	1.32	1.68	1.99	2.27	2.51	2.71	2.90	3.07	3.24	3.38	3.51	3.64	3.77

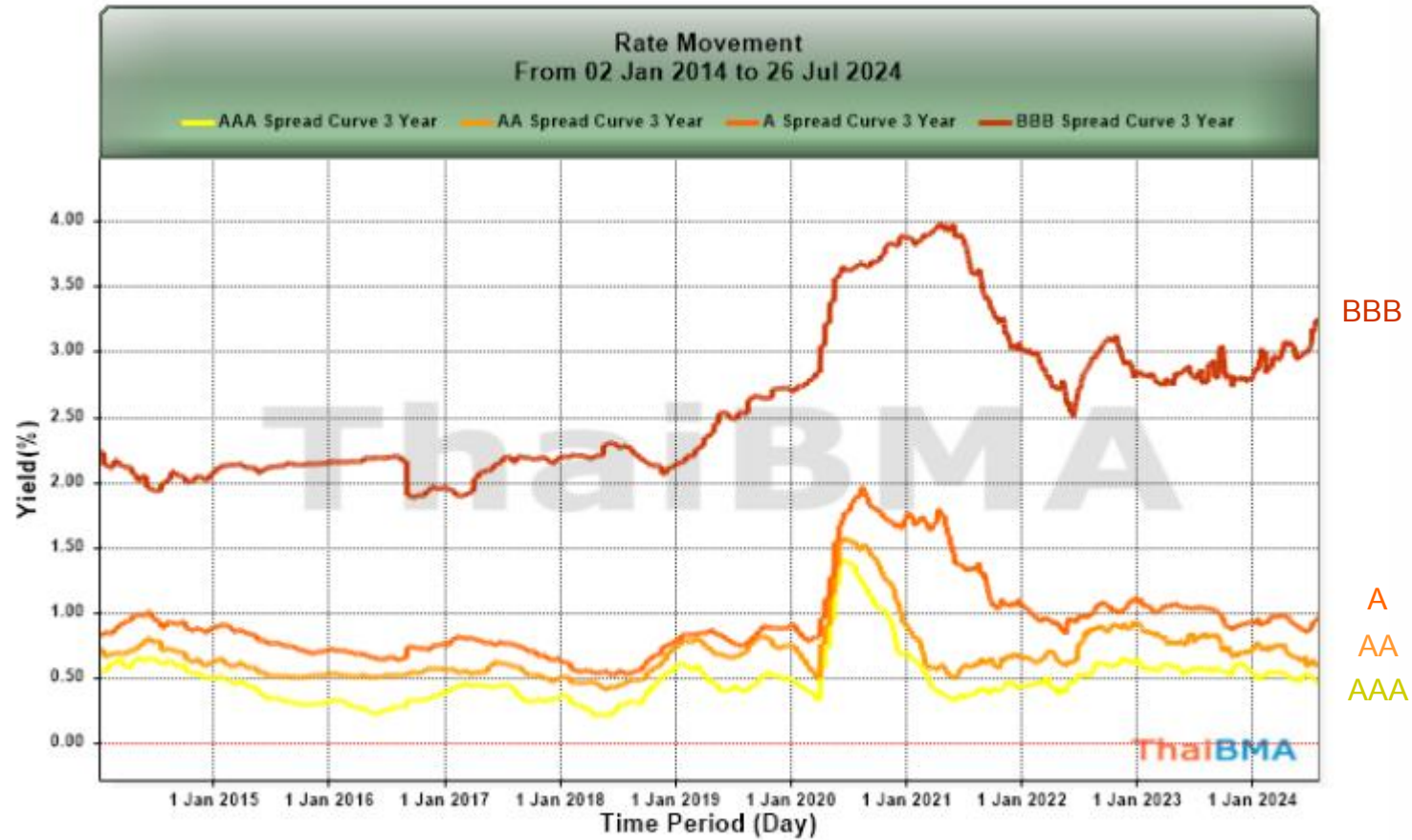
Includes the bank, nonbank financial institution, and insurance sectors. Sources: S&P Global Ratings Credit Research & Insights and S&P Global Market Intelligence's CreditPro®.

Probability of Default Rate for Thai Corporate Bonds

12-month Probability of Default

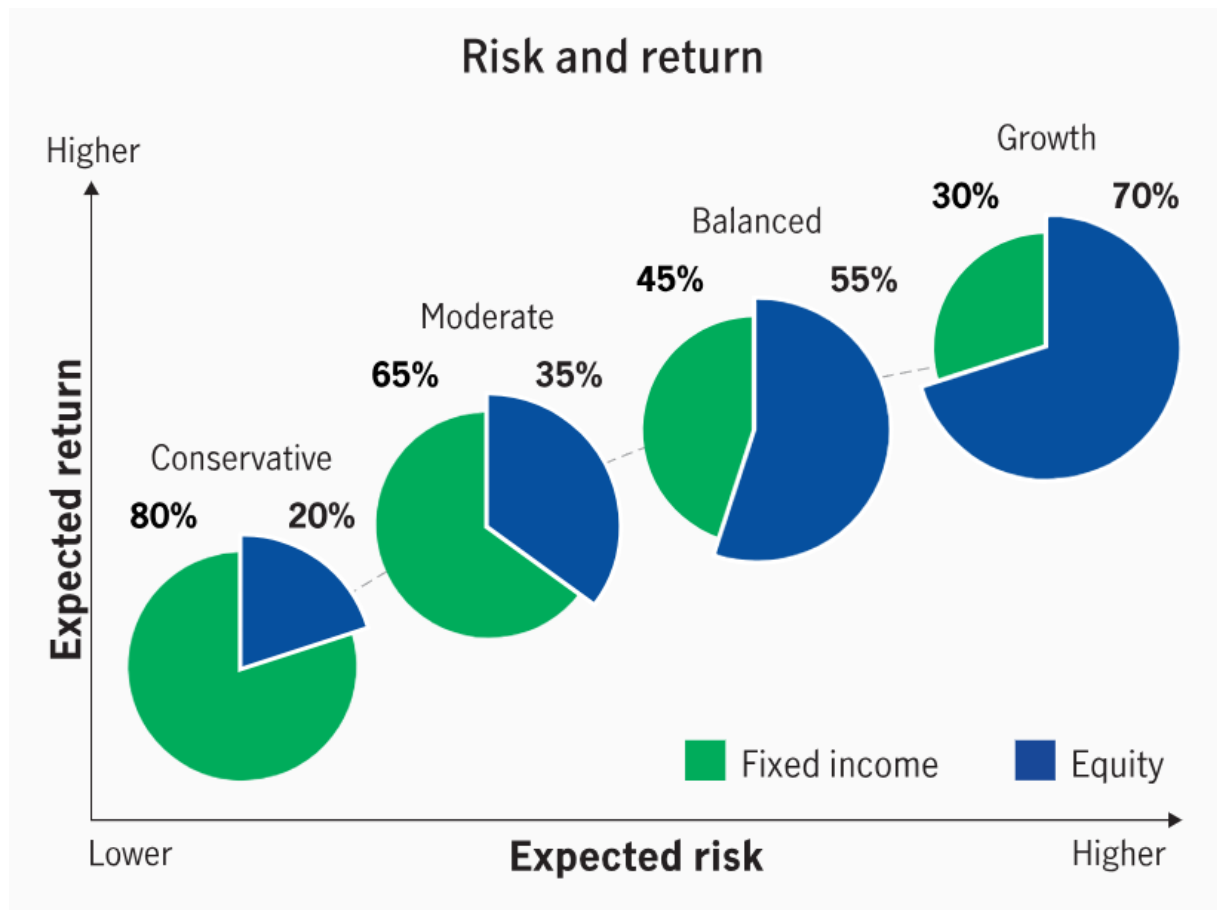
Credit Rating	Probability of Default (%)
AAA	0.013863%
AA+	0.015708%
AA	0.020710%
AA-	0.036311%
A+	0.084336%
A	0.182095%
A-	0.334362%
BBB+	0.581789%
BBB	0.905014%
BBB-	1.260243%
BB+	1.655292%
BB	2.088579%
BB-	2.483153%
B+	2.876818%
B	3.580547%
B-	6.451879%
CCC+	15.463347%
CCC	30.590526%
CCC-	46.314165%
CC	66.565314%
C	87.207708%
Default	100.000000%

Credit Spread Movement of Thai Corporate Bond Market



The Role of Fixed Income in a Portfolio

Example of Multi Asset Portfolio



- **Capital preservation:** Unlike equities, bonds should repay principal at a specified date, or maturity. This makes bonds appealing to investors who do not want to risk losing capital and to those who must meet a liability at a particular time in the future
- **Income:** Most bonds provide the investor with “fixed” income. On a set schedule, whether quarterly, twice a year or annually, the bond issuer sends the bondholder an interest payment
- **Capital appreciation:** Bond prices can rise for several reasons, including a drop in interest rates and an improvement in the credit standing of the issuer.
- **Diversification:** Including bonds in an investment portfolio can help diversify the portfolio.
- **Potential hedge against an economic slowdown or deflation**